LESSON PLAN

AVIONICS AND ELECTRONIC WARFARE (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	22
Total No. of Lesson Plans	25	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Landing, gear control: - Landing gear position indication, anti-skid control systems, wind shield wiper systems, rain repellent systems, air
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Landing, gear control: - Landing gear position indication, anti-skid control systems, wind shield wiper systems, rain repellent systems, air conditioning systems.
5. Specific Objectives	13.1.3.1 Knowledge on circuit protection devices, generators, actuators & engine starting system.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of system schematics and logic diagrams; Safety: Ensure no live electrical connections are simulated; Instructor supervision: Monitor student understanding of circuit paths.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the landing gear position indication system?
(b) Name two types of circuit protection devices used in avionics.
(c) How does an anti-skid system prevent wheel lock-up during landing?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Landing gear position indication systems • Anti-skid control system operation • Windshield wiper system components • Rain repellent system activation • Air conditioning system basics	L & Ds	PPT, IB	15 min
(b)	• Circuit protection devices analysis • Generator and actuator functions • Engine starting system overview • System interconnections review • Troubleshooting common failures	L & Ds	PPT, PDF	20 min
(c)	• Review of specific objectives • Key terminology clarification • System reliability factors • Maintenance requirements • Safety protocols for systems	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the role of circuit protection devices in the engine starting system.
(b) Describe how the anti-skid system interacts with the landing gear control.
(c) What are the primary functions of the rain repellent system?

12. Home Assignment and Follow Up	Prepare a brief note on the differences between warning lights and advisory lights, referencing the next lesson plan (LP 23) on central warning systems.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Warning lights, caution lights, indicating or advisory lights, magnetic indicators, central warning systems, electronic display systems.	Refer to the master syllabus reference list	LP 23

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min